

KUDANKULAM NUCLEAR POWER PLANT



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Located on the coast of the Gulf of Mannar, at 25 KM to the north-east from Kanyakumari, in Tamil Nadu State.

In Kudankulam we use world's most advanced VVER-1000 reactors designed by Russians



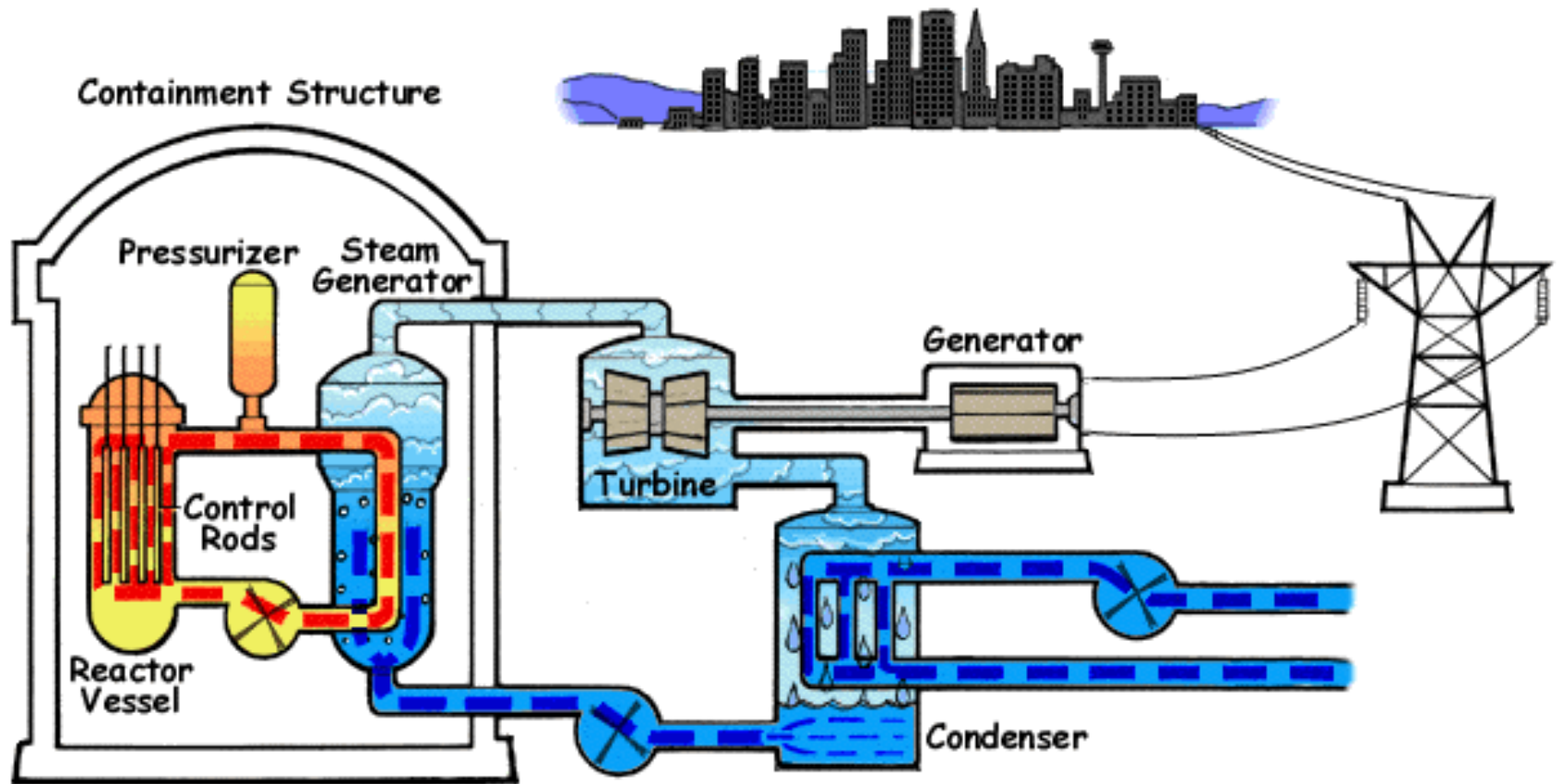
Nuclear Power Plants in India

Power station	Operator	State	Type	Units	Total capacity(MW)
<u>Kaiga</u>	<u>NPCIL</u>	<u>Karnataka</u>	<u>PHWR</u>	220 x 4	880
<u>Kakrapar</u>	<u>NPCIL</u>	<u>Gujarat</u>	<u>PHWR</u>	220 x 2	440
<u>Kalpakkam</u>	<u>NPCIL</u>	<u>Tamil Nadu</u>	<u>PHWR</u>	220 x 2	440
<u>Narora</u>	<u>NPCIL</u>	<u>Uttar Pradesh</u>	<u>PHWR</u>	220 x 2	440
<u>Rawatbhata</u>	<u>NPCIL</u>	<u>Rajasthan</u>	<u>PHWR</u>	100 x 1	1180
				200 x 1	
				220 x 4	
<u>Tarapur</u>	<u>NPCIL</u>	<u>Maharashtra</u>	<u>BWR</u> <u>(PHWR)</u>	160 x 2	1400
				540 x 2	
Total				20	4780

The projects under construction are:

Power station	Operator	State	Type	Units	Total capacity (MW)
<u>Kudankulam</u>	<u>NPCIL</u>	<u>Tamil Nadu</u>	<u>VVER-1000</u>	1000 x 2	2000
<u>Kalpakkam</u>	<u>BHAVINI</u>	<u>Tamil Nadu</u>	<u>PFBR</u>	500 x 1	500
<u>Kakrapar</u>	<u>NPCIL</u>	<u>Gujarat</u>	<u>PHWR</u>	700 x 2	1400
<u>Rawatbhata</u>	<u>NPCIL</u>	<u>Rajasthan</u>	<u>PHWR</u>	700 x 2	1400
Total				7	5300

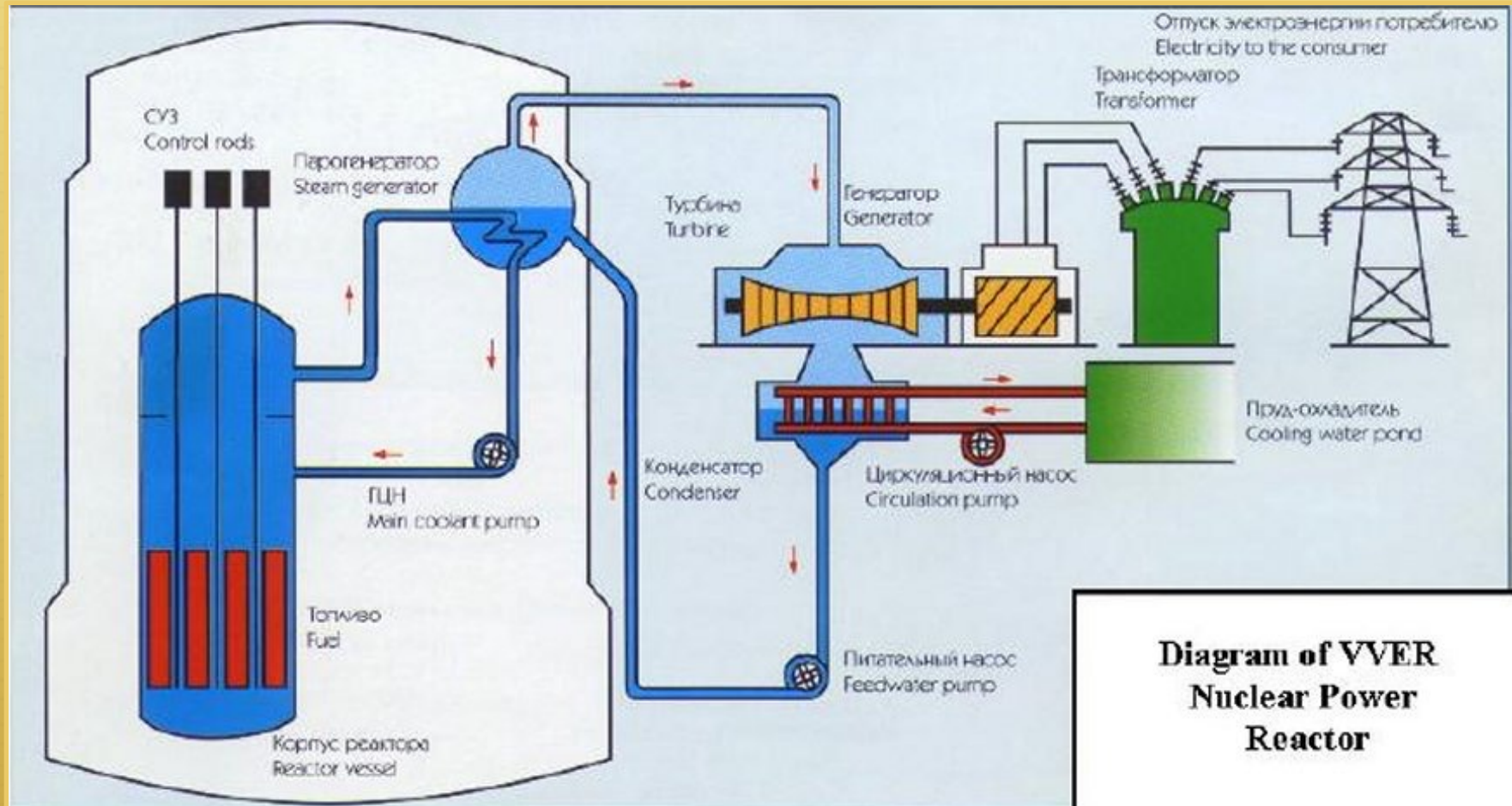
What is a nuclear reactor



What kind of reactor is used in Kudankulam ?

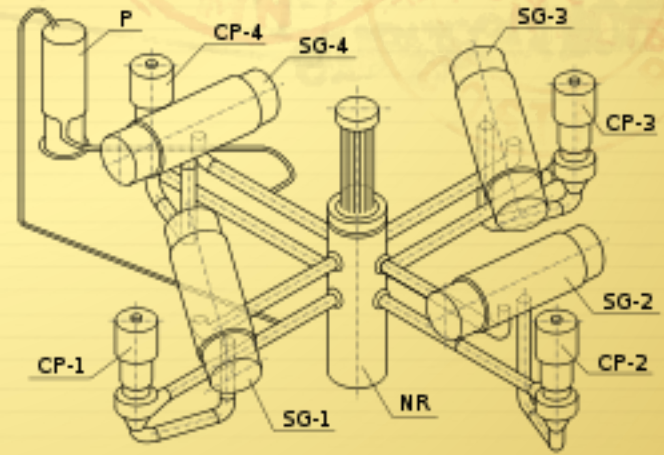
- VVER (WWER)-Vodo-Vodyanoi Energetichesky Reactor (water- water power reactor)
First version → VVER – 440 model V230
Modified version → VVER- 440 model V213 (includes safety standards)
advanced version → VVER-1000 (one used in Kudankulam plant)
- VVER-1000 developed by Russian in 1975, capacity of generating 1000 MW.
 - a) Water cooled, water-moderated- energy reactor (similar to PWR).
Reactor fuel is fully immersed in water at 15 Mpa of pressure.
 - b) Fuel is low enriched (2.4–4.4% ^{235}U) uranium dioxide (UO_2),pressed into pellets and assembled into fuel rods.
 - c) Has a four loop system housed in a containment type structure.

VVER – Russian PWR(pressurized water reactor) (Water-Cooled, Water-Moderated, Energy Reactor)



Primary cooling circuit

- **Reactor** : Water flows through fuel rod assemblies and is heated by the nuclear chain reaction.
- **Volume compensator** : To keep the water under constant but controlled pressure.
- **Steam Generator** : heat from primary coolant water is used to boil water in the secondary circuit.
- **Pump** : The pump ensures proper circulation of the water through the circuit.



Secondary circuit and electrical output

- **Steam Generator** : Secondary water is boiled taking heat from the primary circuit.
- **Turbine** : The expanding steam drives a turbine, which connects to an electrical generator.
- **Condenser** : The steam is cooled and allowed to condense, shedding waste heat into a cooling circuit.
- **Deaerator** : Removes gases from the coolant.
- **Pump** : The circulation pumps are each driven by their own small steam turbine.

Safety barriers

A typical design feature of nuclear reactors is layered safety barriers preventing escape of radioactive material. VVER reactors have four layers:

- **Fuel pellets** : Radioactive elements are retained within the crystal structure of the fuel pellets.
- **Fuel rods** : The zircaloy tubes provide a further barrier resistant to heat and high pressure.
- **Reactor Shell** : A massive steel shell.
- **Reactor Building** : A concrete containment building.

Local protest



SAFETY AT KUDANKULAM

- ✓ Inherent safety –
 1. Reactor has negative power coefficient, which means that any increase in reactor power is self terminating.
 2. Reactor has negative void coefficient, reactor will shut down if there is any loss of water.
- ✓ Reactor is equipped with five barrier to prevent radioactivity into the environment.
 1. Fuel matrix
 2. Fuel cladding
 3. Main circulation circuit
 4. Inner containment system
 5. Outer containment system.
- ✓ Double containment building –
 1. Primary containment designed to maintain pressure at 0.4Mpa
 2. Secondary Containment designed for external effects, such as missile attack, aircraft crash and shockwaves.
- ✓ Two stage Hydro Accumulators – They ensure the supply of borated water to the reactor core in case of loss of coolant or low pressure.
- ✓ Core catcher – in extreme situation of core melting and breaching the pressure vessel, the core catcher below the pressure vessel will ensure that the molten core mixes with a large quantity of neutron absorbing material and thus prevent the possibility of a nuclear explosion. Only the most modern designs have such a provision.

- ✓ The reactor can withstand external effects involving earthquake, tsunami/storm, tidal waves, cyclones, shock waves, fire and aircraft impact.
- ✓ Kudankulam is located far off (about 1500 km) from the tsunami genic fault (where tsunami originate). Thus, if there is a tsunami, it would take time and lose its energy by the time it strikes Kudankulam site.
- ✓ In case of flood - The supplementary control room and the four diesel generator room – safety train rooms are provided with water tight doors to protect them against flooding. These doors remain shut when the reactors are operational.
- ✓ Kudankulam is located in the zone of very low seismic potential.

"Kudankulam plant has 3G safety aspects, I am fully satisfied," Mr Kalam said.

Is it necessary to plant any more nuclear reactors ?

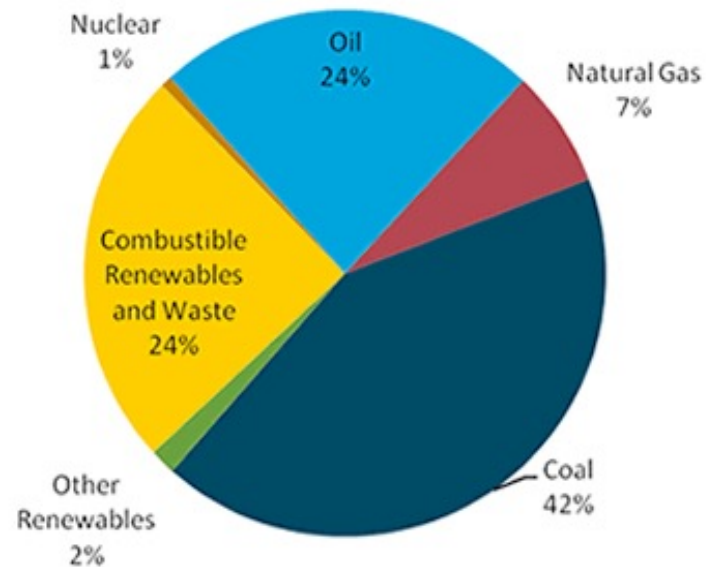
India is the fourth largest energy consumer in the world, after the United States, China, and Russia.

India produced roughly 0.95 million barrels of oil per day. The country consumed 3.2 million barrels per day. So, we import 2.2 million barrels per day from Saudi Arabia and Iran.

Coal based thermal power plant generate 80 percent of electricity in India.

India suffers from shortage of electricity. Roughly 40 percent of residences in India are without electricity.

**Total Energy Consumption in India,
By Type (2009)**



Source: The International Energy Agency

Future of Indian Nuclear Power

- ✓ India's nuclear power reactors consume, at most, 478 tonnes of uranium per year.
- ✓ Even if we quadruple our nuclear power output to 20GW by 2020, nuclear power generation would only consume 2000 tonnes of uranium per annum.
- ✓ India has viable reserves of 80,000 to 112,000 tons of uranium, this represents a 40–50 years uranium supply for India's nuclear power Reactors.
- ✓ India is developing thorium reactor since Uranium deposits in the nation are limited, but there are much greater reserves of thorium and it could provide hundreds of times the energy with the same mass of fuel.
- ✓ **ITER (International Thermonuclear Experimental Reactor)** is an international nuclear fusion research and engineering project, which is currently building the world's largest and most advanced experimental nuclear fusion reactor.
- ✓ ITER's member states — the EU, India, Russia, China, South Korea, Japan and the United States.





Thank you